King Saud University
College of Computer and Information Sciences
Department of Information Technology

IT 462:

Big Data Systems

Course Project

2st semester, 2025/26

Diabetes Health Indicators

Phase 5

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Table of Contents

1. Problem Formulation

The objective of this phase is to develop a machine learning model capable of predicting diabetes status using the Diabetes Health Indicators dataset. The task is formulated as a multi-class classification problem because the target variable contains three categories:

- 0 = No Diabetes
- 1 = Prediabetes
- 2 = Diabetes

The model utilizes 20 health, behavioral, and demographic features including BMI, HighBP, HighChol, PhysActivity, Smoker, GenHlth, Age, Education, and Income. These variables were selected because they are commonly associated with diabetes risk and general health condition.

One important challenge in the dataset is class imbalance, since most records belong to the non-diabetic category. This increases baseline prediction accuracy and makes classification of minority classes more difficult.

2. Choice of Model

The Random Forest Classifier was chosen as the primary analytical framework for this classification task, as it is exceptionally well-suited for the complexities inherent in healthcare and demographic datasets. By leveraging an ensemble learning approach that constructs multiple decision trees, this model demonstrates superior robustness against overfitting in comparison to individual decision trees. Furthermore, it effectively manages the non-linear relationships and intricate variable interactions typically found in medical research. Critically, the Random Forest architecture provides an integrated mechanism to calculate feature importance, which facilitates a transparent interpretation of the specific variables driving the model's predictive output.

3. Evaluation Metrics

The Random Forest model was evaluated using a 70/30 train-test split with a fixed random seed of 42 to ensure reproducibility.

- Accuracy:

The model achieved an overall classification accuracy of 83.49%.

● F1-Score:
The model achieved an F1-score of 78.28%, which provides a more balanced evaluation for imbalanced multi-class classification problems.

● Baseline Comparison:
To evaluate whether the model learned meaningful patterns beyond majority-class prediction, the results were compared with a baseline classifier that always predicts the dominant class. The baseline accuracy was 82.82%.

Although the Random Forest model slightly outperformed the baseline, the improvement was relatively small. This result reflects the strong class imbalance in the dataset, where most records belong to the non-diabetic category. Therefore, accuracy alone may overestimate model effectiveness, while the F1-score provides a more realistic measure of predictive performance across all classes.

4. Interpretation of Key Features and Limitations

Feature importance analysis revealed several strong predictors of diabetes status within the Random Forest model.

- **Dominant Predictors:**

The most influential features identified by the model were:

- General Health (GenHlth): 0.2101
- High Blood Pressure (HighBP): 0.2065
- Body Mass Index (BMI): 0.1610
- High Cholesterol (HighChol): 0.0931
- Difficulty Walking (DiffWalk): 0.0805

These findings are consistent with healthcare expectations, since poor general health, obesity, and high blood pressure are strongly associated with diabetes risk.

- **Lifestyle Variables:**

Lifestyle-related variables such as fruit consumption, vegetable consumption, smoking, and physical activity showed lower feature importance scores compared to physiological indicators.

This suggests that chronic health conditions contributed more strongly to model predictions than isolated behavioral variables.

Error Analysis:

The model performed best on the dominant non-diabetic class, while prediction performance for prediabetes and diabetes classes remained more limited. This behavior is mainly caused by class imbalance, since significantly fewer records belong to minority classes.

Another possible source of error is overlap between classes. Individuals with prediabetes may share similar health characteristics with both healthy and diabetic participants, making separation more difficult for the model.

The lack of clinical laboratory measurements may also reduce prediction accuracy because important diagnostic indicators are unavailable in the dataset.

Limitations:

Despite achieving strong overall accuracy, the model demonstrated only a modest improvement over the majority-class baseline because of significant class imbalance in the dataset. Most survey responses belong to the non-diabetic category, which naturally increases baseline accuracy.

Additionally, the dataset relies on self-reported survey responses, which may contain subjective bias or inaccurate reporting. The absence of clinical biomarkers such as blood glucose measurements or A1C levels also limits the clinical depth of the predictive framework.

Future improvements could include class balancing techniques, hyperparameter tuning, and incorporation of additional clinical features to improve prediction performance for minority classes.